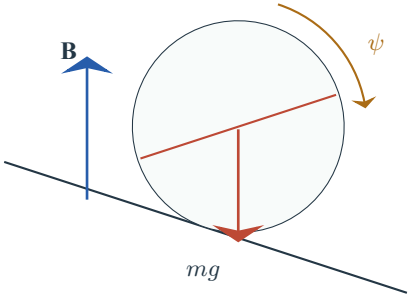
理论图 滚动线框的磁通与力矩



实心圆柱无滑动滚动；$x = r\psi$

$$Li + 2Brh\cos\psi = 2Brh$$

$$i = 2Brh(1 - \cos\psi)/L$$

$$J_{\text{eff}}\ddot{\psi} = mgr\sin\alpha - \frac{4B^2r^2h^2}{L}(1-\cos\psi)\sin\psi$$